



ALL



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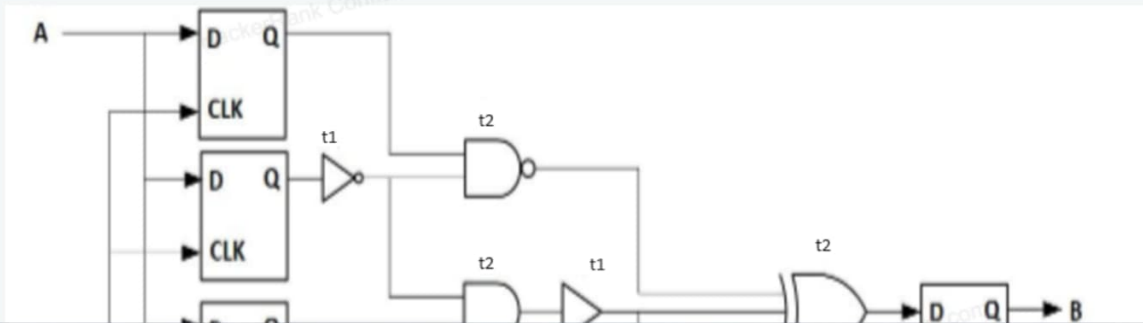


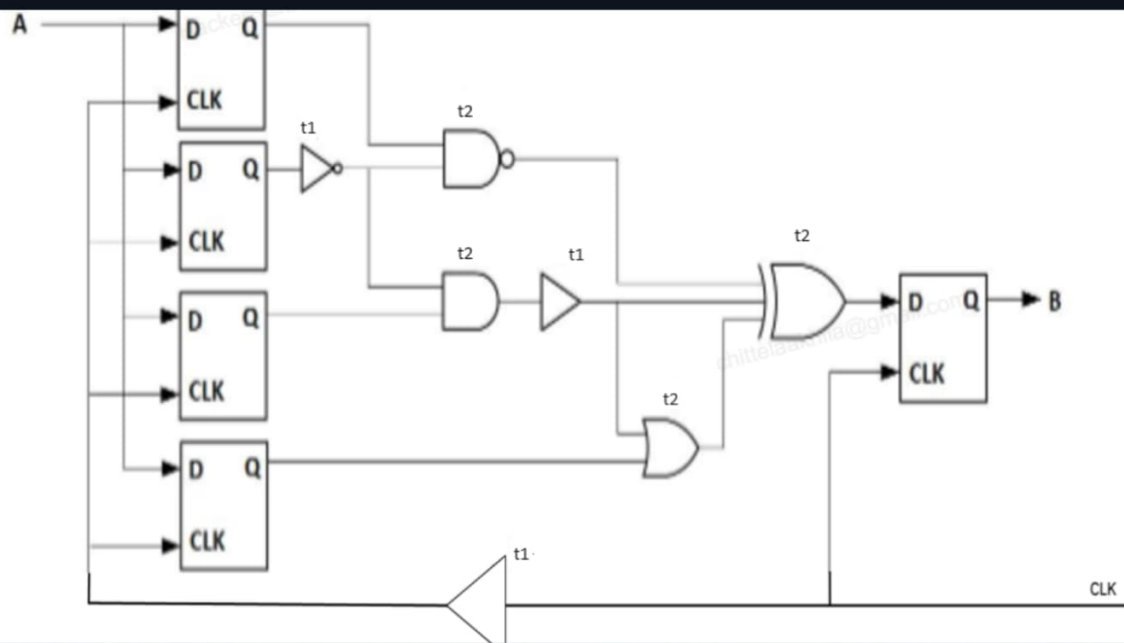
1. Question 1

Complete the blanks in the following question with the appropriate answer.

Find the minimum clock period that is required for operating the following circuit. Please provide the answer in ns.

Note - Setup Time, Hold time & Clock-to-q delays are the same for all flip flops. All single input combinational gates have same delay of t_1 and all multi-input combinational gates have same delay of t_2







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	min delay (ns)	max delay (ns)
t1	3	5
t2	2	6
tck to q	7	9
tsetup	2	
thold	2	

Minimum clock period the circuit can operate is 20 ns



ALL



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5. Question 5

Complete the blanks in the following question with the appropriate answer.

A user uses generative AI (like ChatGPT) to write an essay. The complexity of the essay is determined by the number of words.

For a N word essay, the generative AI activates N GPU nodes in a data center to generate the essay & return the essay back to the user in T_1 seconds and while doing so each GPU node consumes D Watts of dynamic power when active.

If the essay needs to generate pictures along with words, the generative AI scales the operating frequency of all the involved GPU nodes by X factor and scales the voltage of all the GPU nodes by V factor to return the essay quickly to the user in same time.

The data center charges $\$M$ for every kiloWatt-sec of energy consumed.

**GPU DATA
CENTER**



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GENERATIVE AI



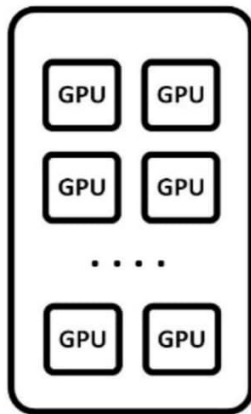
REQUEST



RESPONSE



GPU DATA
CENTER



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Parameter	Value
N	14
X	2
V	3
T1	20
D	50
M	1

1. Total Dynamic power consumed by the data center for a essay comprising only of words is 700 W
2. Total Dynamic power consumed by the data center for a essay comprising of both words & pictures is 2100 W
3. Total cost that data center charges for generating a essay comprising of both words & pictures is \$ 2.1



ALL



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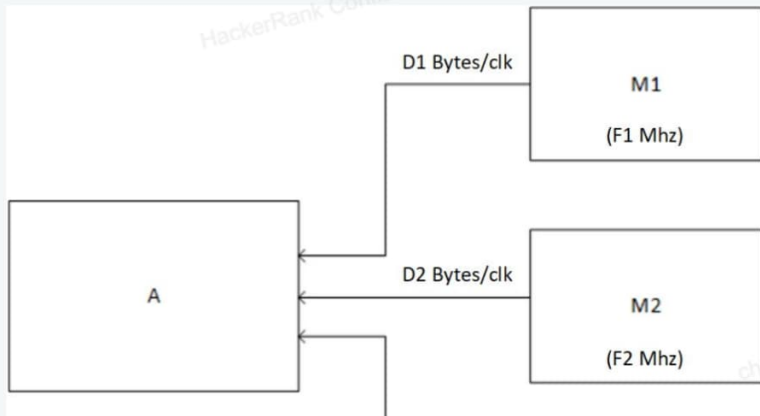
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8

Complete the blanks in the following question with the appropriate answer.

IP A reads data from 3 different memories - Memories M1, M2 & M3.

Memories M1, M2 & M3 communicate with IP A with the data rates & frequency as mentioned in the figure below.





ALL



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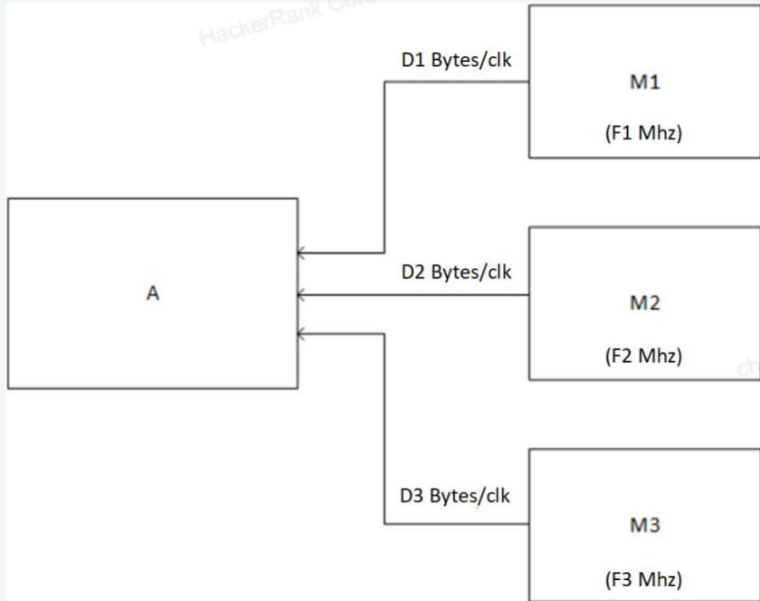


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ALL



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Given the below parameters:

Parameter	Value
D1	1
D2	8
D3	16
F1	5
F2	5
F3	5

Q1. The maximum possible bandwidth is _____ MB/s

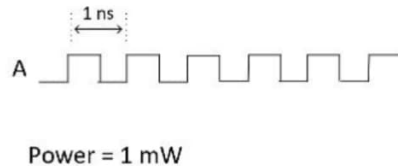
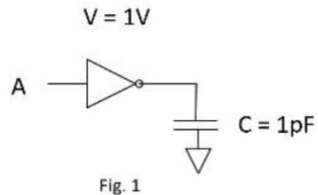
Q2. The total data required by A is stored equally among the 3 Memories. The maximum data read Bandwidth possible for IP A is _____ MB/s

Q3. For achieving the BW in Q1, _____ % of total data should be in M1

7. Question 7

Complete the blanks in the following question with the appropriate answer.

Power dissipation of the circuit in Fig. 1 is 1mW. Assume parasitics are negligible compared to load C. Determine the power for circuit in Fig. 2. $x = 2\text{ V}$, $y = 0.5\text{ pF}$.



$V = 1V$

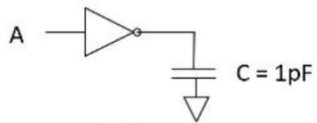


Fig. 1



Power = 1 mW

$V = x V$

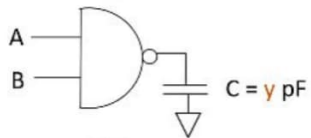
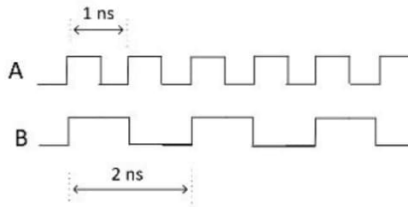
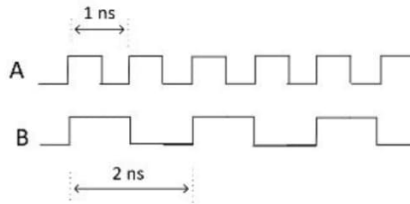
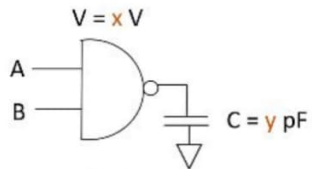


Fig. 2



Power = ?



Power = ?

For circuit in Fig. 2, Power = _____ mW



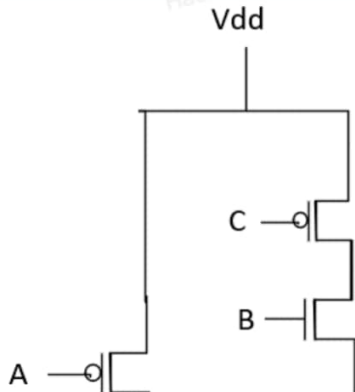
Complete the blanks in the following question with the appropriate answer.

ALL



Find voltage at node Y.

$V_t = 0.2 \text{ V}$, $V_{dd} = 1 \text{ V}$





ALL



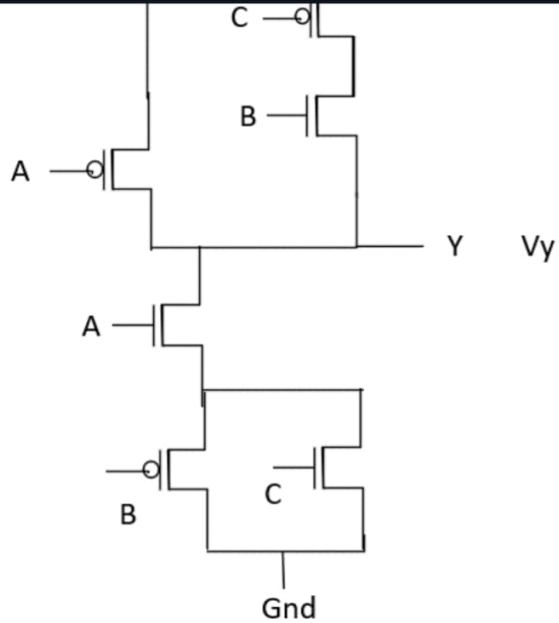
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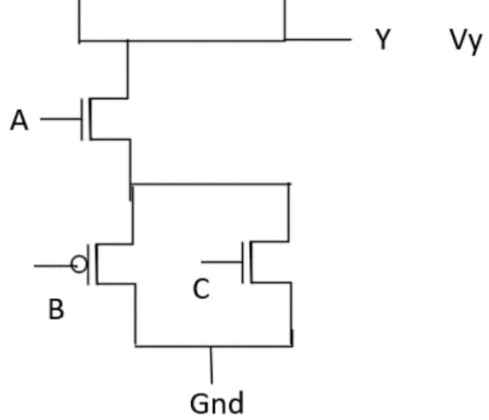
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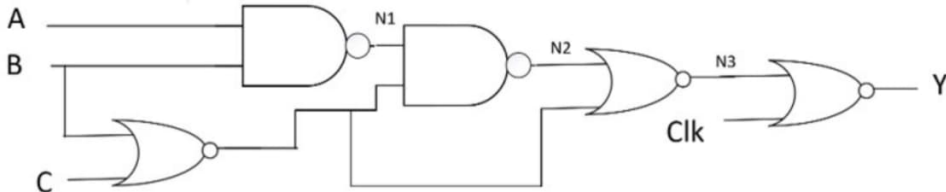


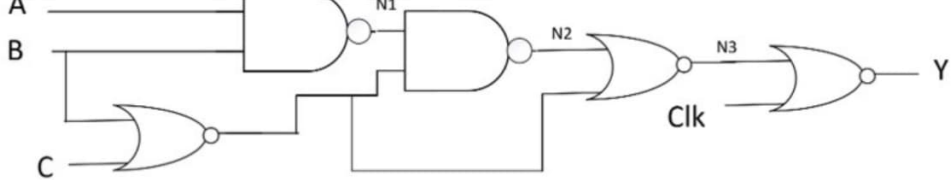
when $A = 1$, $B = 1$, $C = 0$, $V_y =$ _____ V

9. Question 9

Complete the blanks in the following question with the appropriate answer.

TNOR Rise = 50 ps, TNOR Fall = 40 ps, TNAND Rise = 45 ps, TNAND Fall = 38 ps





Setup time of pins

A = _____ ps,

B = _____ ps,

C = _____ ps.



ALL



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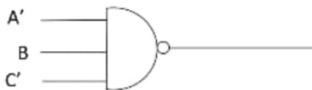
10. Question 10

Complete the blanks in the following question with the appropriate answer.

Evaluate the logical expression for this circuit.

Note :

1. Use ' to represent complement value. For eg: If answer is NOT of X, then please write it as X'
 2. Use logical operators "|" and "&" for OR and AND resp.
 3. Use parentheses "(" ")" to group symbols.
 4. Don't use any spaces between the variables or logical operators.
- For eg : If answer is "(M AND N) OR (NOT of P) ", then please write it as (M&N)|P'



ALL

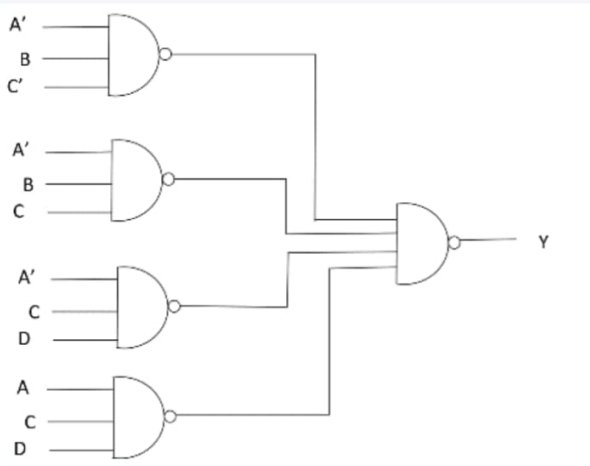


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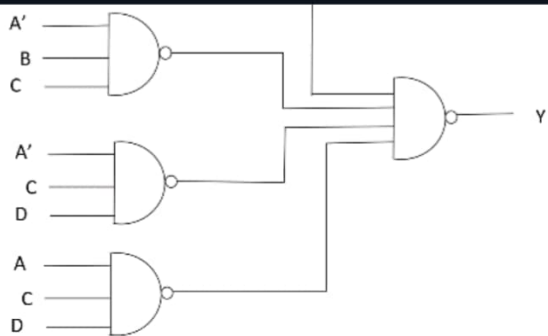
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Y= _____



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11. Question 11

Complete the blanks in the following question with the appropriate answer.

A Double Data Rate Synchronous Dynamic Random-Access Memory (DRAM) supports a bandwidth of B Gbps(Giga bits per second). The data bus width for the DRAM is X bit.

B	8192
X	2048

The DRAM clock frequency required to support full DRAM bandwidth is _____ GHz

Submit Answer & Continue



ALL



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Complete the blanks in the following question with the appropriate answer.

A display screen is arranged as rows and columns of pixels. A screen has R rows and C columns of pixels. Each pixel needs X bits of storage - for Red, Green, Blue and Alpha (transparency) components. F represent the Frame Rate Per Second. N represent the compression Rate in Memory. S represent the size of SRAM

R	256
C	512
X	64
F	16
N	2
S	3



ALL



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X	64
F	16
N	2
S	3

Total size of memory required to store one frame of video is _____ MB (MegaBytes)

Note if 1.545 or 1.546 round to 1.55 . If 1.563 round off to 1.56

Bandwidth required to stream video at F frames per second is _____ MBPS (MegaBytes Per Second)

A compressor can compress the frame by a factor of N:1. No. of compressed frames that can be fit in a SRAM of S MB is _____



ALL



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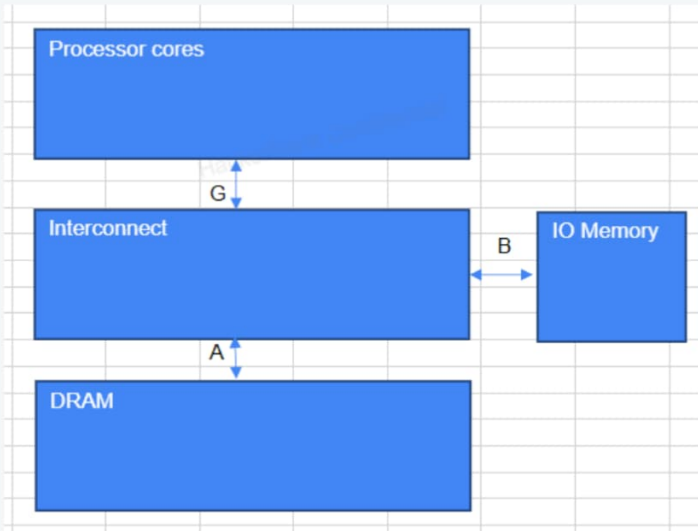
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Complete the blanks in the following question with the appropriate answer.



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ALL



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A GPU has a bunch of processor cores (SMs) connected to a high speed DRAM and high speed IO Memory. The DRAM and IO Memory are connected to the SMs through an Interconnect. The DRAM has a data bandwidth of A GBPs. The IO Memory has a data bandwidth of B GBPs. The Interconnects makes use of a protocol that has efficiency of C %.

A	768
B	256
C	80%

Total Raw bandwidth(Data bandwidth + Inefficiency) that SMs needs to handle including protocol inefficiencies is _____ GBPs

The interconnect has several ports, each of which can carry 32 GBPs. Number of interconnect ports needed to provide adequate total bandwidth to the processor cores is _____



ALL



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Complete the blanks in the following question with the appropriate answer.



A cuda program has 4 phases P1, P2, P3 and P4. P1 is limited by memory latency, P2 is limited by floating point throughput, P3 is limited by memory bandwidth and P4 is limited by IO bandwidth. The percentage of time spent in each phase on the current generation of GPU is given below. You are an engineer who is tasked with speeding up this program on the next gen GPU architecture.

% of time spent in P1	20%
% of time spent in P2	50%
% of time spent in P3	15%
% of time spent in P4	15%



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% of time spent in P1	20%
% of time spent in P2	50%
% of time spent in P3	15%
% of time spent in P4	15%

You have found a way to increase Memory bandwidth by "M" times, IO bandwidth by "I" times, Floating point throughput by "F" times, while Decreasing the memory latency by "L" times in the next generation of GPU

L	1
F	2
M	2
L	2



ALL



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You have found a way to increase Memory bandwidth by "M" times, IO bandwidth by "I" times, Floating point throughput by "F" times, while Decreasing the memory latency by "L" times in the next generation of GPU

L	1
F	2
M	2
L	2

The Program Speedup by _____.

Assume that the bottleneck doesn't shift in the new generation of GPU as well.

Note if 1.545 or 1.546 round to 1.55 . If 1.563 round off to 1.56



15. Question 15

ALL



Complete the blanks in the following question with the appropriate answer.

Find the output of the following program given

M = {1,2,3,4,5,6,7}

N = {4,5,6}

```
using v = std::vector<int>;  
  
#define L(k) ((k?k:new v())->size())  
  
int foo(v* s = NULL) {  
    static v* x = NULL;  
    v* y = NULL;  
    static int r;  
  
    s?x=s,y=s:s;  
  
    r += L(x);  
}
```



ALL



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```
using v = std::vector<int>;

#define L(k) ((k?k:new v())->size())

int foo(v* s = NULL) {
    static v* x = NULL;
    v* y = NULL;
    static int r;

    s?x=s,y=s:s;

    r += L(x);
    r -= L(y);

    return r;
}

int main() {
    v vec1 = M;
    v vec2 = N;

    foo(&vec1); foo();
    foo(&vec2); foo();

    std::cout << "answer: " << foo() << std::endl;

    return 0;
}
```

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ALL



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```
    r += L(x);  
    r -= L(y);  
  
    return r;  
}  
  
int main() {  
    v vec1 = M;  
    v vec2 = N;  
  
    foo(&vec1); foo();  
    foo(&vec2); foo();  
  
    std::cout << "answer: " << foo() << std::endl;  
  
    return 0;  
}
```

Answer: _____



ALL



Complete the blanks in the following question with the appropriate answer.

Find the output of the following program given

X = "NVIDIAN"

```
int ans = 0;
void foo(string s)
{
    if (s.length() < 1) {
        ans++;
    }
    unordered_set<char> uset;
    for (int i = 0; i < s.length(); i++) {
        if (uset.find(s[i]) != uset.end()) {
            continue;
        }
    }
}
```




ALL



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```
int ans = 0;
void foo(string s)
{
    if (s.length() < 1) {
        ans++;
    }
    unordered_set<char> uset;
    for (int i = 0; i < s.length(); i++) {
        if (uset.find(s[i]) != uset.end()) {
            continue;
        }
        else {
            uset.insert(s[i]);
        }
        string temp = "";
        if (i < s.length() - 1) {
            temp = s.substr(0, i) + s.substr(i + 1);
        }
        else {
            temp = s.substr(0, i);
        }
    }
}
```

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```
    }  
    else {  
        uset.insert(s[i]);  
    }  
    string temp = "";  
    if (i < s.length() - 1) {  
        temp = s.substr(0, i) + s.substr(i + 1);  
    }  
    else {  
        temp = s.substr(0, i);  
    }  
    foo(temp);  
}  
}  
int main()  
{  
    string s = X;  
    foo(s);  
    cout << ans << endl;  
    return 0;  
}
```



ALL



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```
        else {  
            temp = s.substr(0, i);  
        }  
        foo(temp);  
    }  
}  
  
int main()  
{  
    string s = X;  
    foo(s);  
    cout << ans << endl;  
    return 0;  
}
```

Answer: _____



ALL



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17. Question 17

Complete the blanks in the following question with the appropriate answer.

What is the output of the following program assuming

int is 32-bits wide

A = 2

B = 9

```
int main()
{
    int a = A;
    int b = B;

    int *pv = (int*) malloc(10 * sizeof(int));
    for(int i=0, j=0; i<10; i++, j+=a) {
        pv[i] = j;
```

What is the output of the following program assuming

int is 32-bits wide

A = 2

B = 9

```
int main()
{
    int a = A;
    int b = B;

    int *pv = (int*) malloc(10 * sizeof(int));
    for(int i=0, j=0; i<10; i++, j+=a) {
        pv[i] = j;
    }
    printf("%0d", *(pv-a+b));
    return 0;
}
```



ALL



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18. Question 18

Complete the blanks in the following question with the appropriate answer.

What is the height of the binary search tree with the following preorder traversal

Preorder: 10,9,8,1,4,2,3,7,5,6

Height = _____

Submit Answer & Continue



Complete the blanks in the following question with the appropriate answer.

ALL



In the given circuit, four D flip-flops are arranged with the input of the first flip-flop (D_0) derived from a combinatorial circuit using the outputs of the other flip-flops. Initially, a clear signal is applied to the flip-flops. Subsequently, clocks are provided to each flip-flop, causing Q_0 , Q_1 , Q_2 , and Q_3 to assume different values. Assuming a start value denoted as S ($Q_0Q_1Q_2Q_3$) and a final value denoted as F ($Q_0Q_1Q_2Q_3$), determine the minimum number of clock cycles required to transition from the start value to the final value.

Consider

S 0000

F 0010

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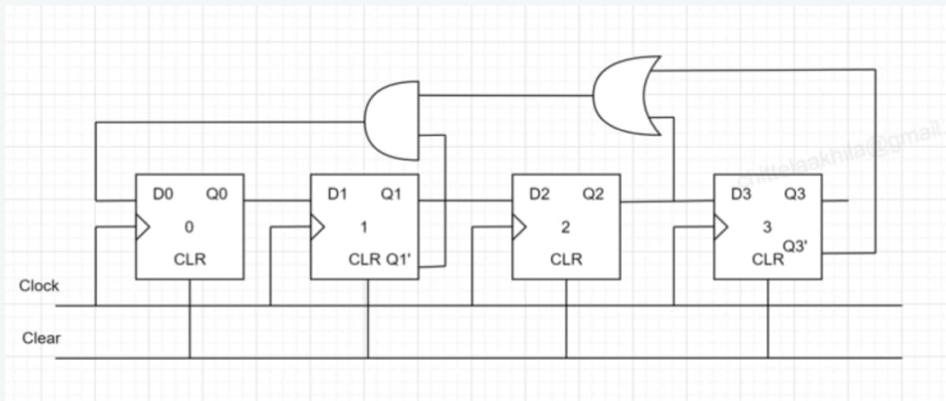
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Number of minimum clocks (in decimal) to reach the final state : _____



ALL



2

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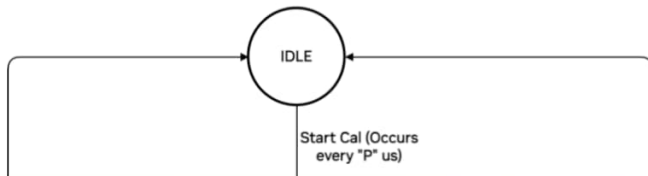
3. Question 3

Complete the blanks in the following question with the appropriate answer.

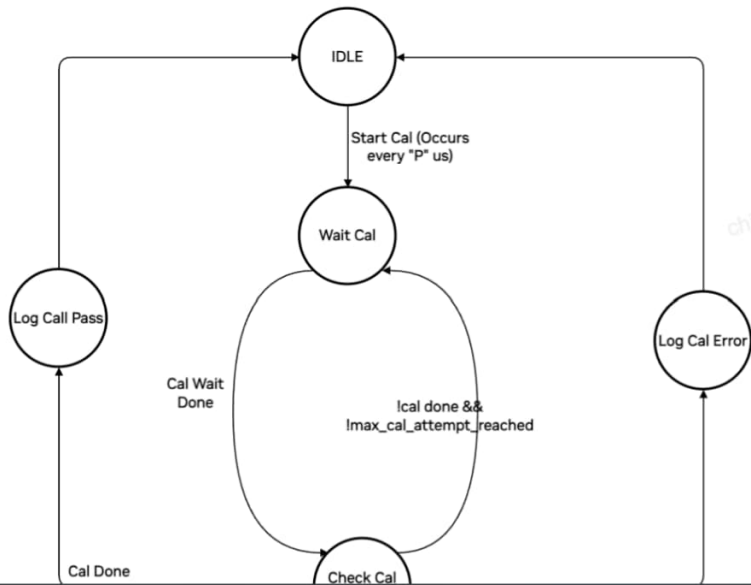
A temperature sensitive sensor is calibrated periodically using the FSM shown in the figure below to account for changes in chip temperature during functional mode.

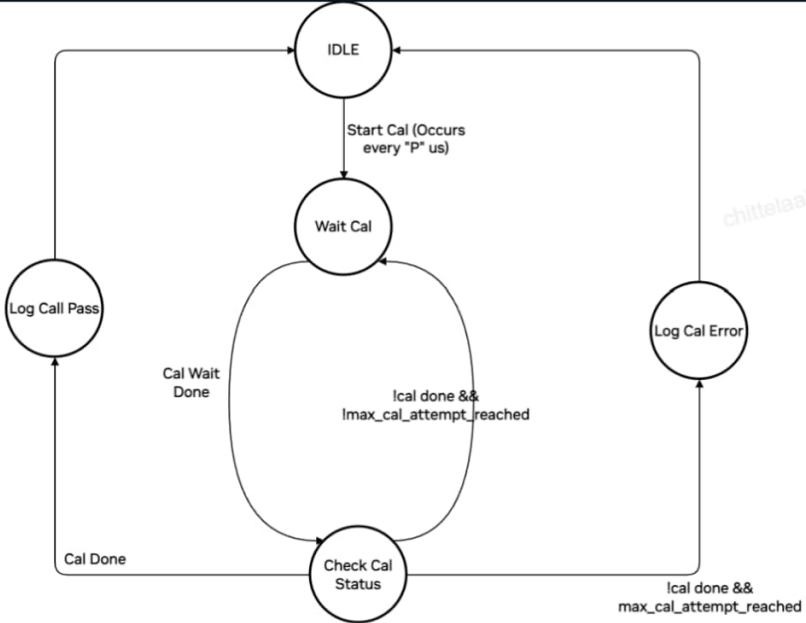
In IDLE state, a counter (width N) is used to generate periodic pulse to start CAL (as shown in the figure). Once cal is started the FSM will poll for cal_done status every "**O**" us. If cal_done is detected, then cal_status is logged as PASS. Otherwise, it tries for a max number of attempts ("**A**"), then cal_status is logged as ERROR. Also the sensor calibration time depends on the temperature of the chip. Below nominal temp ("**T**" degrees), cal_done can take "**H**" us. For every **1 degree** rise in temp, it takes "**h**" us incrementally.

FSM runs at frequency of "F" Mhz



FSM runs at frequency of "F" Mhz







ALL



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P	200
F	20
O	5
A	20
T	40
H	10
h	2

Width of the counter required to generate the periodic Start CAL pulse is _____

Threshold temperature after which the CAL will start to report errors (round up to nearest whole number) is _____



ALL



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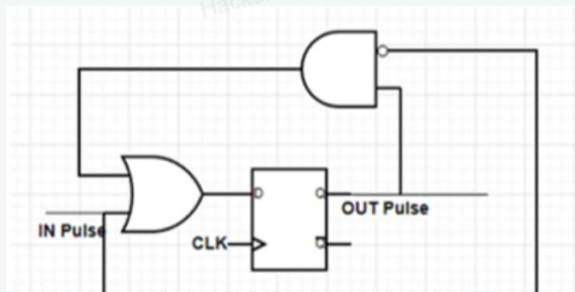
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4. Question 4

Complete the blanks in the following question with the appropriate answer.

Following digital circuit is designed to stretch an input pulse (IN), to generate an output pulse (OUT). The counter is under reset as long as the IN pulse is high, and starts counting up once the IN pulse is de-asserted.

For a given IN pulse width "**Win**", and CLK Frequency **M**. What should be the min. width of the counter to generate an OUT pulse which is at least as wide as "**Wout**"





ALL



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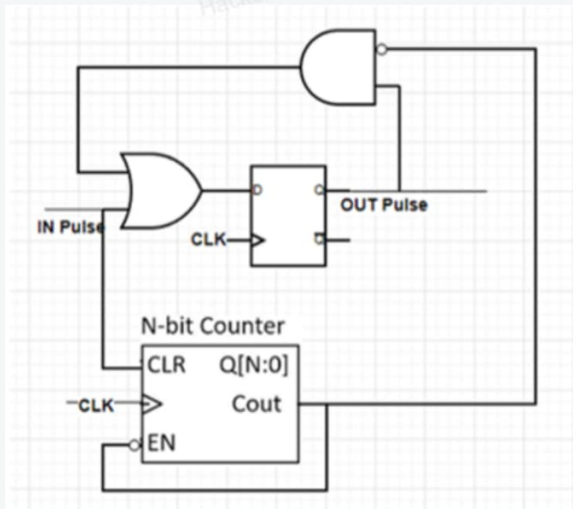
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For a given IN pulse width T_{IN} and CLK frequency f_{CLK} , what should be the min width of the counter to generate an OUT pulse which is at least as wide as "Wout"





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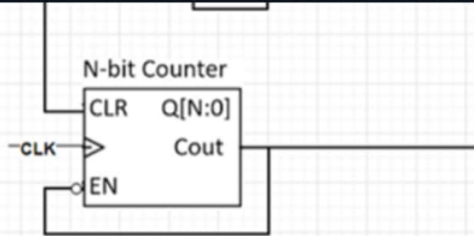
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M (in MHz)	200
Win (in ns)	500
Wout (in us)	15

Minimum width of the counter required to generate the OUT Pulse is _____